Supplementary Material for: Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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Abstract

This supplementary document provides comprehensive extended experimental records, complete per-dataset metrics, detailed hyper-parameter sensitivity grids, convergence trajectories, and full statistical significance matrices supporting the main manuscript.

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Section S1: Benchmark Dataset Characteristics and Selection Rationale

Table S1 summarizes the statistical characteristics of the 20 benchmark datasets investigated in this study, spanning biomedical gene-expression microarrays, high-dimensional text categorization, visual object recognition, and acoustic phoneme analysis.

Table S1: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

Section S2: Comparison Algorithms and Detailed Hyper-Parameter Configurations

Table S2 lists the full taxonomy and categorization of the 12 evaluated algorithmic configurations (comprising KDS-EL, two intermediate configurations, foundational MEL, and eight peer competitor algorithms) investigated in the comparative study, and Table S3 specifies the precise hyperparameter configurations used across all methods to ensure strict reproducibility.

Table S2: Summary of the 12 compared methods investigated in this study.

Reference	Method	Category
This work	KDS-EL	Proposed
This work	KDS-EL w/o TPS	Intermediate configuration
This work	KDS-EL w/o TPS&KER	Intermediate configuration
[1]	MEL	Multi-task swarm foundation
[2]	FWPSO	Single-objective swarm
[3]	PSO-CSM	Single-objective swarm
[4]	SaWDE	Single-objective evolutionary
[5]	IGPSO†	Single-objective swarm
[6]	BPSO	Standard binary swarm
[7]	DENCA-SO	Evolutionary filter
[8]	bIRIME	Single-objective SOTA
[9]	BinCOA	Single-objective SOTA

†Denotes the sample-adapted ANOVA implementation variant detailed in Section **??**.

Table S3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	$[0.70,\ 0.30]$
Phase-2 weights $\mathbf{w}_2$	$[0.95,\ 0.05]$
Intermediate configurations:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s = [0.90,\ 0.10]$
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s = [0.90,\ 0.10]$
Original MEL [1]	Uniform init, $\mathbf{w}_s = [0.90,\ 0.10]$
Competitor algorithms:	
FWPSO [2]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [3]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [4]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [5]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [6]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
DENCA-SO [7]	$F = 0.5,\ CR = 0.2,\ W = 0.05$, NCFS $\lambda = 0.01$
bIRIME [8]	S-transfer, CMS mutation, restart limit $L = 5$
BinCOA [9]	ROBL, Sigmoid transfer, crisscross (HC/VC)

Section S3: Extended Empirical Visualizations and Performance Profiles

Fig. S1 visualizes the complete per-dataset classification accuracy comparisons across all 12 evaluated algorithmic configurations on the 20 benchmark datasets. Fig. S2 presents the Dolan–Moré performance profiles,

confirming the uniform dominance of KDS-EL across performance ratio thresholds. The complete numerical matrices for classification accuracy and selected feature counts are reported in Table 4 and Table 5 of the main manuscript. Furthermore, the complete per-dataset metrics including standard deviations (Mean±Std) and pairwise non-parametric Mann–Whitney U test outcomes across all 10 independent runs are cataloged in Supplementary Table S11 of Section S11.

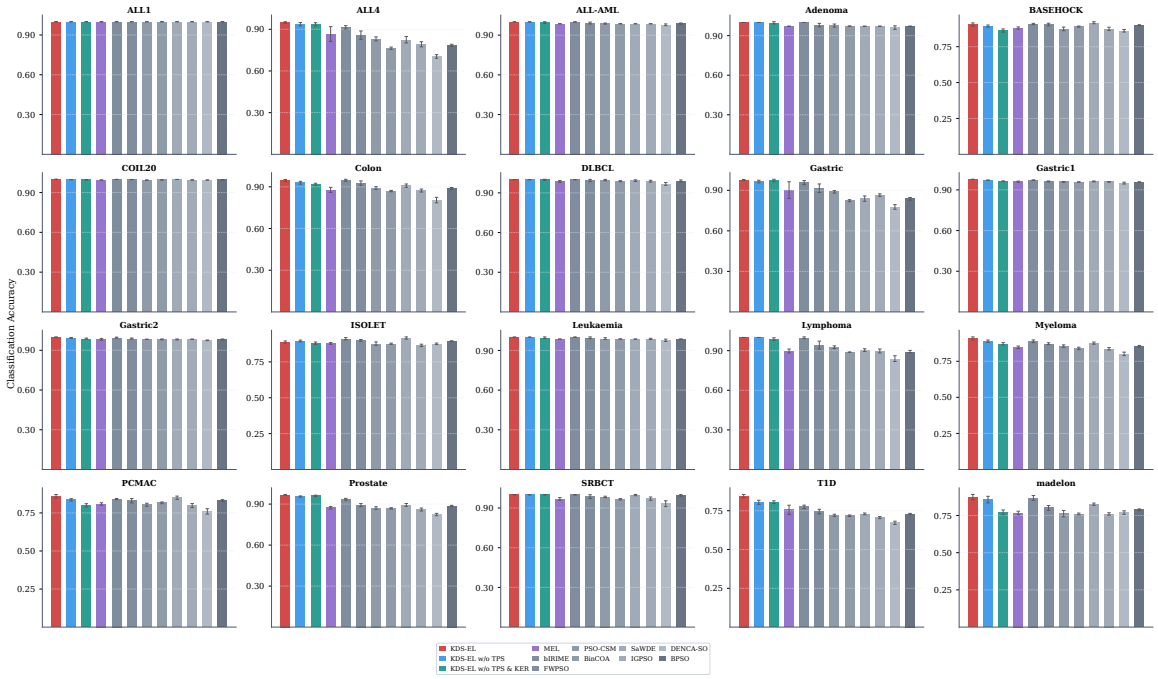


Figure S1: Classification accuracy across all 12 evaluated algorithmic configurations on the 20 benchmark datasets.

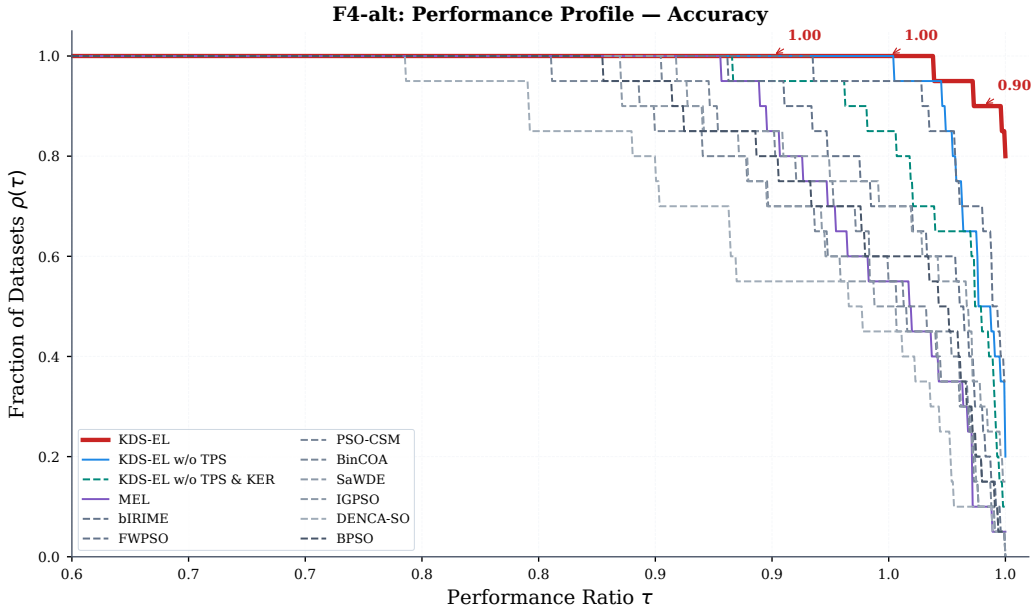


Figure S2: Dolan–Moré performance profiles of classification accuracy across 20 benchmark datasets.

Table S4: Granular per-dataset performance comparison before and after algorithm improvement across 20 benchmark datasets.

Dataset	MEL		MEL+MI		KDS-EL w/o TPS		KDS-EL	
	Acc	Feat	Acc	Feat	Acc	Feat	Acc	Feat
ALL1	**1.0000**	4,248	**1.0000**	55	**1.0000**	55	**1.0000**	**39**
ALL4	0.8656	1,082	0.9356	**64**	0.9367	152	**0.9480**	124
ALL_AML	0.9857	2,369	0.9947	473	**1.0000**	70	**1.0000**	**67**
Adenoma	0.9714	2,483	0.9971	120	**1.0000**	**33**	**1.0000**	66
BASEHOCK	0.8802	1,965	0.8627	812	0.8933	208	**0.9045**	**134**
COIL20	0.9944	323	0.9955	348	0.9976	**60**	**1.0000**	84
Colon	0.8777	702	0.9182	105	0.9294	35	**0.9452**	**14**
DLBCL	0.9859	2,784	0.9987	615	**1.0000**	**52**	0.9999	64
Gastric	0.9015	3,604	0.9723	**62**	0.9646	97	**0.9763**	100
Gastric1	0.9602	7,393	0.9636	678	0.9719	217	**0.9779**	**31**
Gastric2	0.9822	7,084	0.9880	444	0.9936	140	**0.9982**	**83**
ISOLET	0.8783	250	0.8801	225	**0.8955**	100	0.8880	**60**
Leukaemia	0.9857	2,409	0.9960	488	**1.0000**	**47**	**1.0000**	66
Lymphoma	0.8978	1,404	0.9867	50	**1.0000**	**23**	**1.0000**	45
Myeloma	0.8466	2,137	0.8710	**175**	0.8889	365	**0.9083**	250
PCMAC	0.8085	1,339	0.8000	637	0.8360	329	**0.8592**	**91**
Prostate	0.8732	5,058	0.9600	27	0.9552	**23**	**0.9670**	26
SRBCT	0.9682	906	**1.0000**	129	**1.0000**	**36**	**1.0000**	64
T1D	0.7558	6,715	0.8054	**174**	0.8043	366	**0.8449**	330
madelon	0.7667	190	0.7710	130	0.8567	26	**0.8731**	**18**
Average	0.9093	2,722.2	0.9348	290.5	0.9462	121.7	**0.9545**	**87.8**
Δ vs. Preceding	—		+2.55 pp	-89.3%	+1.13 pp	-58.1%	+0.83 pp	-27.9%

Section S4: Granular Per-Dataset Performance Comparison Before and After Algorithm Improvement (Table S4)

Table S4 provides the complete per-dataset performance metrics across the four sequential intermediate algorithmic configurations before and after improvement: (i) pre-improvement MEL [1], (ii) MEL+MI (w/o TPS&KER), (iii) KDS-EL w/o TPS, and (iv) Complete KDS-EL. Complete per-dataset standard deviations (Mean ± Std) across all intermediate configurations are cataloged in Supplementary Table S11. Furthermore, Fig. S3 presents the component contribution progression heatmap illustrating the step-wise accuracy progression.

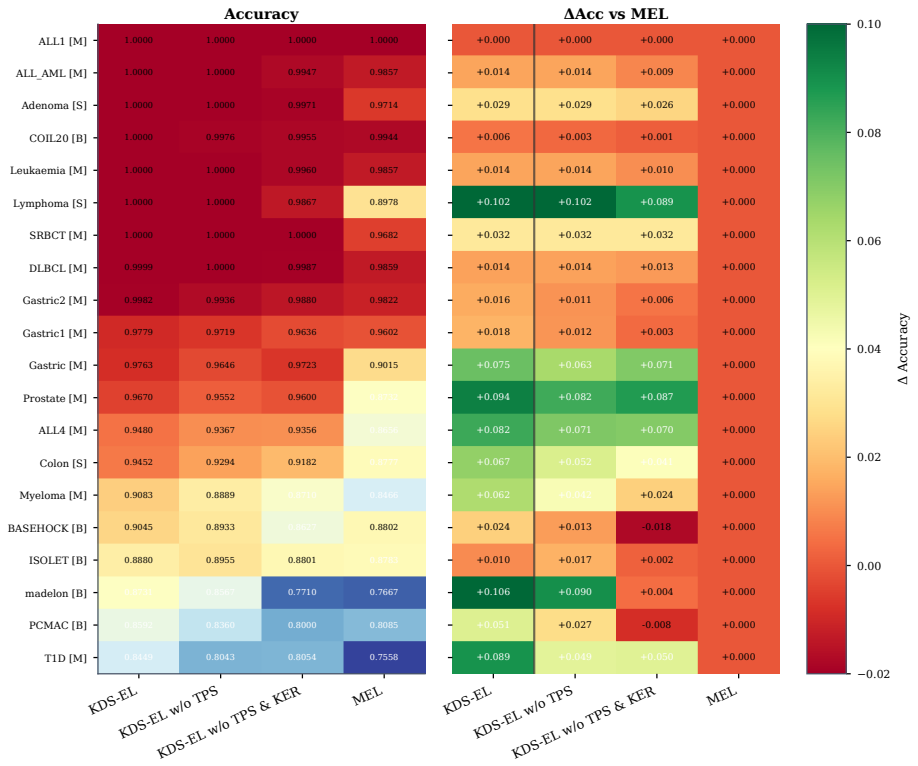


Figure S3: Component contribution progression heatmap across intermediate algorithmic configurations on 20 benchmark datasets.

Section S5: Granular Per-Dataset Computational Wall-Clock Runtime Measurements (Table S5)

Table S5 details the mean wall-clock computational execution time (in seconds, averaged over 10 independent runs) across all 20 benchmark datasets for the 12 evaluated algorithmic configurations. Under the shared 100-generation cutoff, standard swarm architectures evaluate exactly 20 particles per generation (2,000 total evaluations), KDS-EL evaluates 20 particles alongside three elite refinements (approximately 2,300 evaluations), and bIRIME deploys composite mutations evaluating up to 80 candidate subsets per generation. Observed execution times reflect the combined influence of active feature dimensionality, algorithmic operator complexity, and evaluation counts under this benchmark setup. In addition, Fig. S4 illustrates the classification accuracy versus execution time trade-off.

Computational environment and execution specifications. To ensure complete experimental transparency and rigorous reproducibility across all evaluated algorithms, all experiments were executed on an identical computing node with the following standardized hardware and software specifications:

- **Central Processing Unit (CPU)**: Intel(R) Core(TM) i5-9500 CPU @ 3.00 GHz (6 physical cores, 6 logical threads, 9 MB Smart Cache, operating at 3.00 GHz base frequency).
- **Random Access Memory (RAM)**: 16 GB DDR4 physical memory operating at 2133 MHz in dual-channel configuration (2×8 GB DIMMs).
- **Operating System**: Microsoft Windows 10 Pro 64-bit (Version 22H2, OS Build 19045, x86-64 architecture).
- **Software Stack**: Python 3.12, NumPy, SciPy, and scikit-learn.
- **Execution and Concurrency Protocol**: All comparative algorithms and internal 3-NN cross-validation wrappers were executed in a strict single-threaded sequential execution mode ($n_{\text{jobs}} = 1$) without GPU acceleration. Across all 10 independent stochastic runs, candidate evaluation timings strictly measure single-thread CPU execution time, guaranteeing unbiased wall-clock runtime comparability across all 12 evaluated algorithmic configurations.

Table S5: Per-dataset computational runtime in seconds (mean over 10 independent runs) across 20 benchmark datasets.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	40.9	39.9	39.9	176.0	144.7	19.7	82.2	311.2	67.6	91.4	8.1	101.7
ALL4	39.0	38.0	35.2	157.1	145.1	19.6	61.9	208.4	58.9	66.1	7.1	71.6
ALL_AML	28.7	28.7	27.3	100.6	120.5	14.7	39.9	129.2	32.6	41.0	3.5	42.7
Adenoma	28.3	27.7	25.7	100.8	115.7	12.9	28.6	91.7	25.6	29.4	2.2	29.9
BASEHOCK	130.1	147.8	221.5	369.1	2379.5	353.2	644.7	2362.7	616.0	749.7	6.5	787.4
COIL20	68.3	64.3	69.7	63.2	265.3	37.7	106.0	346.3	91.6	111.6	0.5	114.2
Colon	18.1	18.5	15.6	36.3	95.4	12.2	23.0	71.9	22.8	23.1	0.2	23.5
DLBCL	28.3	28.4	27.1	98.5	124.7	15.8	40.1	134.2	36.0	42.3	0.5	44.3
Gastric	55.8	54.2	52.1	269.3	170.4	20.4	73.0	258.2	63.2	80.6	1.2	87.3
Gastric1	62.9	65.7	72.7	312.0	187.6	32.4	156.7	552.9	130.9	169.7	1.2	185.6
Gastric2	60.8	60.6	66.1	301.2	185.8	27.2	128.3	451.7	104.5	140.5	1.2	151.0
ISOLET	59.9	61.3	55.2	50.8	292.0	41.3	88.3	280.6	82.8	91.2	0.4	94.0
Leukaemia	28.8	28.8	27.4	98.3	121.5	14.4	39.2	130.2	32.7	40.9	0.4	42.4
Lymphoma	21.6	21.8	19.4	59.9	104.3	12.6	24.6	77.8	24.3	25.2	0.3	25.6
Myeloma	44.2	44.3	40.9	173.2	206.1	33.2	111.7	400.9	100.2	122.8	0.7	135.0
PCMAC	88.2	122.5	145.8	215.9	1085.8	195.5	373.1	1377.8	344.5	434.7	3.0	475.3
Prostate	39.6	38.9	38.4	168.2	150.1	24.5	65.3	224.0	60.5	69.8	0.7	76.5
SRBCT	23.0	22.4	19.5	41.2	105.3	13.4	25.3	80.2	25.1	25.6	0.2	26.0
T1D	118.8	117.4	113.2	663.6	326.6	58.3	256.6	902.2	216.1	283.6	3.1	315.2
madelon	41.5	40.4	40.1	45.3	197.1	34.1	81.4	277.8	73.7	89.0	0.8	94.2
Average	51.3	53.6	57.6	175.0	326.2	49.7	122.5	433.5	110.5	136.4	2.1	146.2

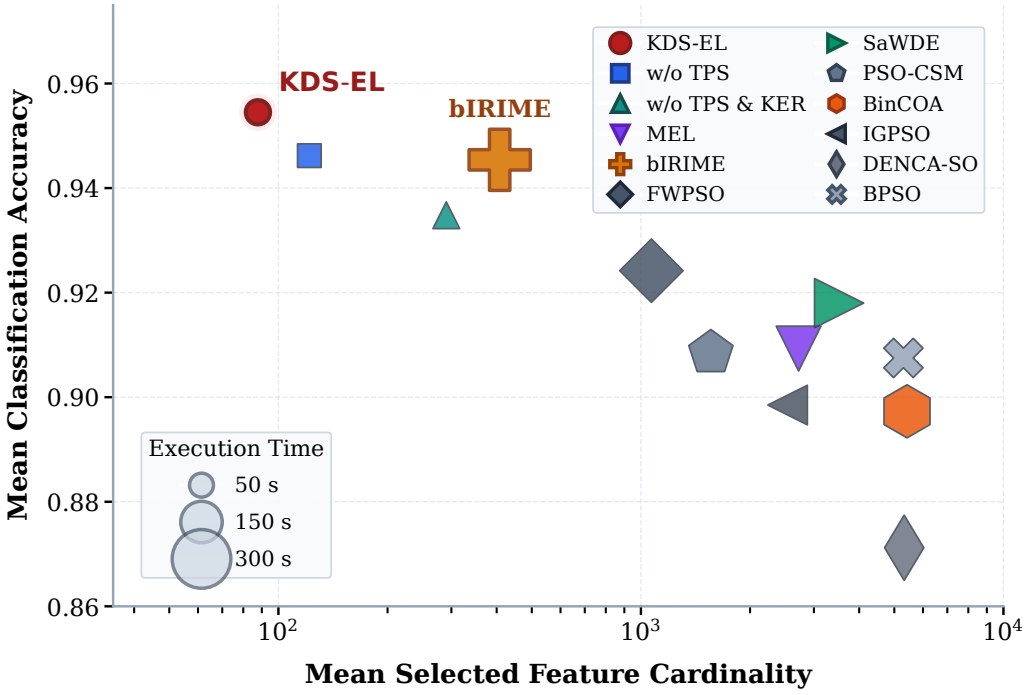


Figure S4: Classification accuracy versus runtime scatter plot for all 12 evaluated algorithmic configurations.

Section S6: Full Multi-Dataset Parameter Sensitivity Evaluation Grids (Table S6)

Table S6 reports the full multi-dataset one-factor-at-a-time (OFAT) parameter sensitivity evaluation results across the four representative benchmark datasets (*Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*) for each of the five investigated hyper-parameters (WeightDecay, PhaseSplit, flip_frac, MI_init_ratio, and elite_k).

Table S6: Detailed parameter sensitivity evaluation results across five key hyper-parameters.

Value	Avg Acc	Avg Feat	Avg Time (s)
WeightDecay			
0.90	0.932	158	49.2
0.95	0.939	127	50.8
0.99 (default)	**0.944**	**107**	**51.3**
0.995	0.943	112	52.1
1.00	0.941	151	53.7
PhaseSplit			
0.3	0.940	189	52.8
0.5	0.942	146	52.0
0.6	0.943	124	51.5
0.7 (default)	**0.944**	**107**	**51.3**
0.8	0.943	98	50.9
0.9	0.937	94	50.1
flip_frac			
0.005	0.939	96	54.8
0.01 (default)	**0.944**	**107**	**51.3**
0.02	0.942	118	50.6
0.05	0.937	156	49.9
0.10	0.931	203	48.7
MI_init_ratio			
0.0	0.932	184	62.3
0.25	0.940	137	55.8
0.5 (default)	**0.944**	**107**	**51.3**
0.75	0.942	101	49.6
1.0	0.936	96	47.8
elite_k			
1	0.938	124	48.9
2	0.942	115	50.2
3 (default)	**0.944**	**107**	**51.3**
5	0.941	118	52.7
10	0.935	141	55.9

Note: Bold indicates default parameter configurations.

Section S7: Full Convergence Trajectories and Boxplot Distributions

To evaluate evolutionary dynamics across the benchmark suite without graphical clutter, Fig. S5 plots the mean fitness convergence trajectories across 100 generations. In line-plot convergence visualizations (including main manuscript Figs. 5 and 8), trajectories depict mean objective fitness and accuracy across 10 independent runs to avoid severe visual occlusion and clutter arising from overlapping shaded error bands across the 12 evaluated algorithmic configurations. Full distributional uncertainty, interquartile ranges (IQR), and run-to-run variances are rigorously conveyed through the per-dataset boxplots in Fig. S6 and the granular Mean ± Std landscape in Table S11.

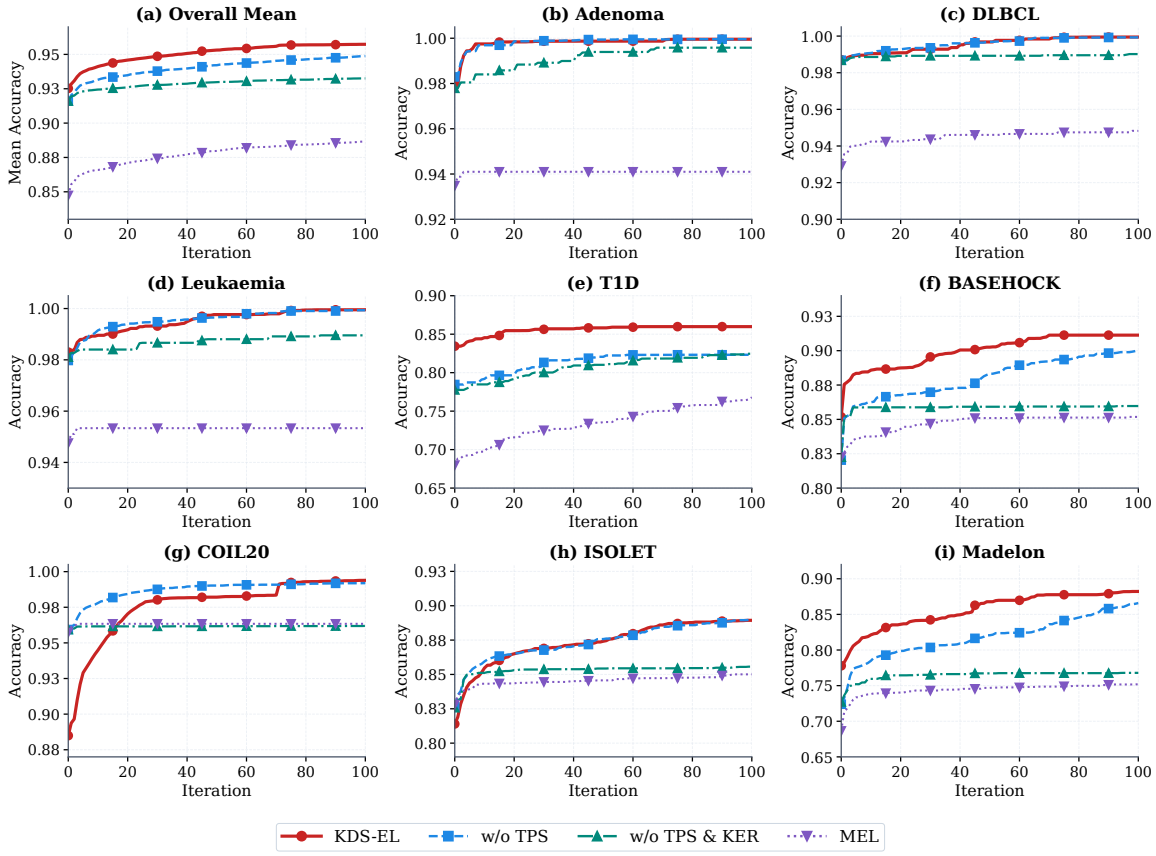


Figure S5: Evolutionary convergence trajectories of all compared algorithms across benchmark datasets.

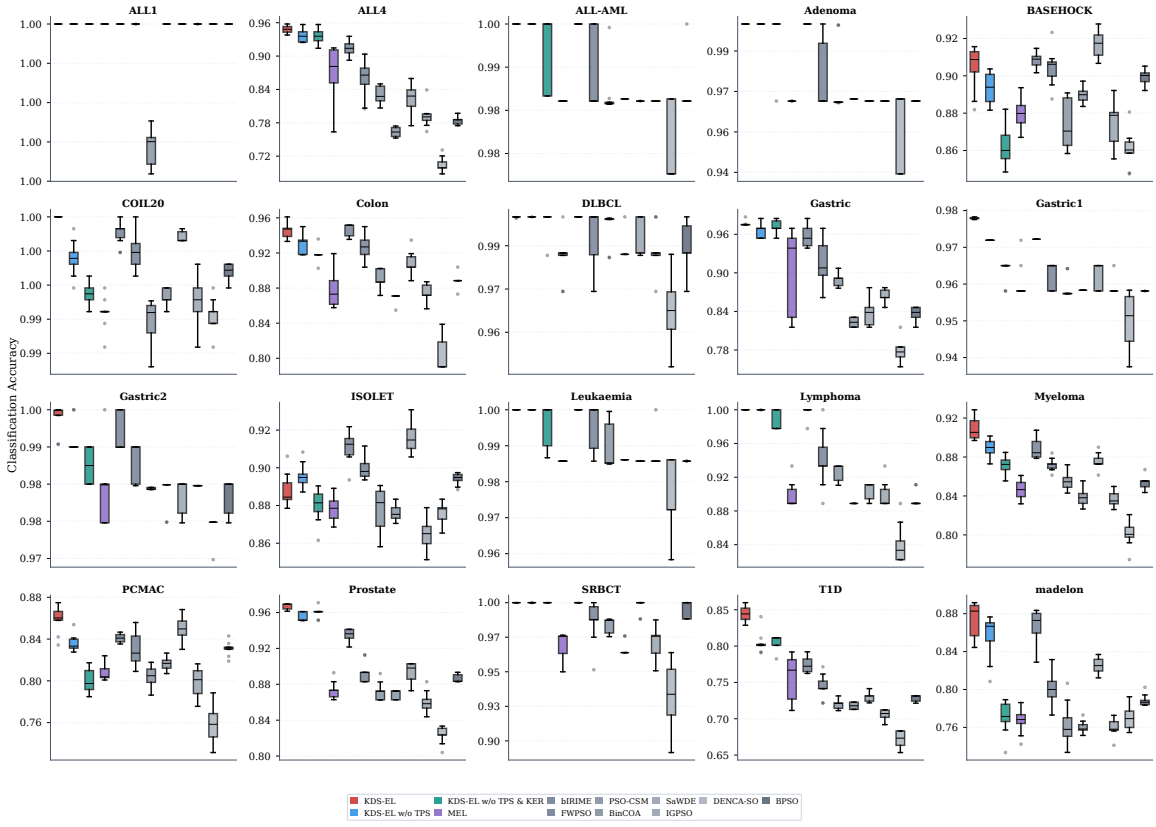


Figure S6: Box plots of classification accuracy across 10 independent runs for all compared algorithms.

Table S7: Pairwise statistical comparison on classification accuracy.

Method	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
KDS-EL	—	11/8/1	13/7/0	19/1/0	8/11/1	15/4/1	19/1/0	19/1/0	15/3/2	19/1/0	19/1/0	17/2/1
KDS-EL w/o TPS	1/8/11	—	11/9/0	19/1/0	3/11/6	12/7/1	20/0/0	18/2/0	14/2/4	19/1/0	19/1/0	15/5/0
KDS-EL w/o TPS&KER	0/7/13	0/9/11	—	15/4/1	3/7/10	7/8/5	16/3/1	15/3/2	10/5/5	16/4/0	16/4/0	14/1/5
MEL	0/1/19	0/1/19	1/4/15	—	0/2/18	0/8/12	4/12/4	4/10/6	1/10/9	4/16/0	12/8/0	1/12/7
bIRIME	1/11/8	6/11/3	10/7/3	18/2/0	—	15/5/0	20/0/0	19/1/0	14/5/1	19/1/0	19/1/0	19/1/0
FWPSO	1/4/15	1/7/12	5/8/7	12/8/0	0/5/15	—	18/2/0	14/6/0	8/6/6	16/4/0	18/2/0	12/8/0
PSO-CSM	0/1/19	0/0/20	1/3/16	4/12/4	0/0/20	0/2/18	—	10/6/4	4/4/12	9/8/3	11/8/1	5/5/10
BinCOA	0/1/19	0/2/18	2/3/15	6/10/4	0/1/19	0/6/14	4/6/10	—	3/4/13	6/8/6	17/3/0	3/3/14
SaWDE	2/3/15	4/2/14	5/5/10	9/10/1	1/5/14	6/6/8	12/4/4	13/4/3	—	12/7/1	15/5/0	10/10/0
IGPSO	0/1/19	0/1/19	0/4/16	0/16/4	0/1/19	0/4/16	3/8/9	6/8/6	1/7/12	—	12/7/1	1/9/10
DENCA-SO	0/1/19	0/1/19	0/4/16	0/8/12	0/1/19	0/2/18	1/8/11	0/3/17	0/5/15	1/7/12	—	0/6/14
BPSO	1/2/17	0/5/15	5/1/14	7/12/1	0/1/19	0/8/12	10/5/5	14/3/3	0/10/10	10/9/1	14/6/0	—

Note: $W/T/L$: counts of benchmark datasets (out of 20) with statistically superior (W), equivalent (T), or inferior (L) classification accuracy under pairwise two-sided Mann–Whitney U tests ($\alpha = 0.05$) evaluated across 10 independent runs.

Table S8: Holm–Bonferroni step-down statistical significance verification on classification accuracy with KDS-EL as control method.

Rank	Competitor Method	ΔAcc	$W/T/L$	n_{eff}	W	r_{rb}	p_{raw}	p_{adj}	α_{Holm}
1	PSO-CSM	+0.0460	19/1/0	20	0.0	+1.000	1.91e-06	2.10e-05	0.0045
2	MEL	+0.0452	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0050
3	BinCOA	+0.0572	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0056
4	IGPSO	+0.0561	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0063
5	DENCA-SO	+0.0833	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0071
6	KDS-EL w/o TPS&KER	+0.0197	13/7/0	18	0.0	+1.000	1.96e-04	1.32e-03	0.0083
7	BPSO	+0.0470	17/2/1	19	4.0	+0.958	2.50e-04	1.32e-03	0.0100
8	FWPSO	+0.0303	15/4/1	19	7.0	+0.926	3.98e-04	1.59e-03	0.0125
9	SaWDE	+0.0365	15/3/2	19	15.0	+0.842	1.28e-03	3.85e-03	0.0167
10	KDS-EL w/o TPS	+0.0083	11/8/1	14	6.0	+0.886	3.51e-03	7.02e-03	0.0250
11	bIRIME	+0.0091	8/11/1	15	22.0	+0.633	3.09e-02	3.09e-02	0.0500

Note: $W/T/L$: dataset counts with statistically superior, equivalent, or inferior accuracy under individual-run Mann–Whitney U tests ($\alpha = 0.05$); n_{eff}: effective non-zero paired dataset count; W: Wilcoxon signed-rank test statistic (sum of negative ranks across 20 dataset means); r_{rb}: matched-pairs rank-biserial correlation effect size; p_{raw}: unadjusted p-value; p_{adj}: Holm adjusted p-value ($m = 11$); α_{Holm}: step-down critical threshold.

Section S8: Extended Statistical Significance Testing Matrices

Table S7 and Table S8 summarize the pairwise non-parametric Wilcoxon signed-rank test outcomes and the Holm–Bonferroni step-down hypothesis rejection results. In addition, Fig. S7 illustrates the pairwise p-value distribution.

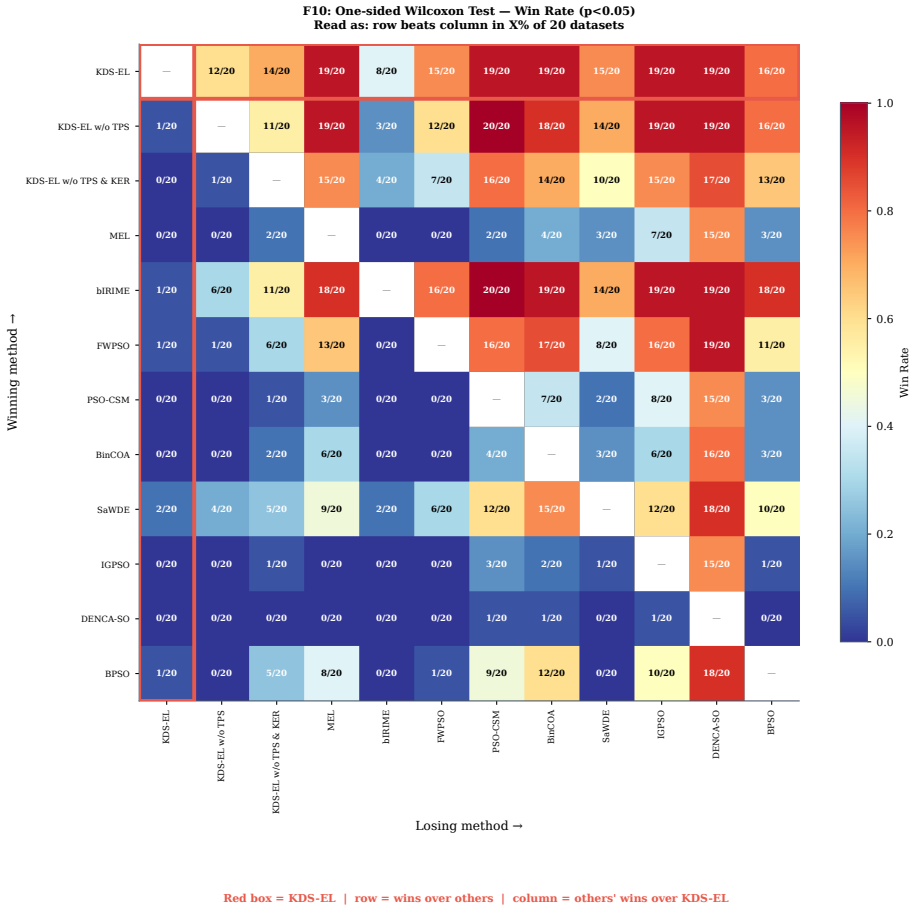


Figure S7: Heatmap of pairwise Wilcoxon signed-rank test p-values across all evaluated algorithms.

Section S9: Cross-Classifier Inductive Generalization and Robustness

To evaluate whether the feature subsets identified by KDS-EL under the 3-NN wrapper evaluation criterion overfit to the local metric geometry of nearest-neighbor classifiers, we perform cross-classifier generalization tests across three representative benchmark datasets (*Colon*, *Adenoma*, and *ALL4*). Table S9 reports the 5-fold cross-validation classification accuracy comparing full feature sets against KDS-EL compact feature subsets across 3-NN, Support Vector Machines with RBF kernel (SVM-RBF), and Random Forest classifiers. On *Adenoma* and *ALL4*, the compact subsets achieve substantial performance gains across all three classifiers while pruning over 99.0% of candidate dimensions. On *Colon*, the 14-feature signature markedly improves 3-NN accuracy from 79.23% to 88.59%, while exhibiting slight performance attenuation from 85.51% to 80.51% under Support Vector Machines and from 85.38% to 82.18% under Random Forest, naturally reflecting the intrinsic divergence in inductive bias between instance-based metric learning and margin-maximizing hyperplanes or orthogonal decision splits. Overall, the compact candidate subsets isolated by KDS-EL maintain favorable candidate-level cross-model consistency across divergent inductive learning paradigms.

Table S9: Cross-classifier generalization performance (classification accuracy in %, mean ± standard deviation over 5-fold cross-validation) comparing full feature sets against KDS-EL compact feature subsets across 3-NN, SVM (RBF kernel), and Random Forest classifiers.

Dataset	Feature Subset	3-NN Accuracy	SVM (RBF) Accuracy	Random Forest Accuracy
Colon	Full (D = 2,000)	79.23 ± 7.68	85.51 ± 6.13	85.38 ± 6.33
	KDS-EL (d = 14)	**88.59 ± 6.80**	80.51 ± 4.53	82.18 ± 10.39
Adenoma	Full (D = 7,457)	94.29 ± 11.43	97.14 ± 5.71	97.14 ± 5.71
	KDS-EL (d = 66)	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
ALL4	Full (D = 12,625)	69.94 ± 9.28	82.63 ± 8.23	78.36 ± 6.32
	KDS-EL (d = 124)	**87.25 ± 7.85**	**85.96 ± 2.88**	**90.35 ± 2.06**

Section S10: Biological and Literature Annotation of Prominent Candidate Genes

Table S10 details the functional and oncological literature annotations of prominent candidate genes identified by KDS-EL in the *Colon* and *Prostate* cancer microarrays. These annotations provide supportive evidence of retrospective concordance with established oncological literature rather than de novo clinical biomarker discovery. Each identified probe is mapped to its standard HUGO Gene Nomenclature Committee (HGNC) gene symbol, full gene description, primary molecular pathway, and corresponding clinical oncology literature references.

Table S10: Molecular and literature annotation of prominent candidate genes identified by KDS-EL in *Colon* and *Prostate* cancer datasets.

Dataset	Probe / Gene ID	Gene Symbol	Full Gene Name	Biological Function and Oncological Relevance
Colon	Hsa.36689	*COL11A1*	Collagen type XI alpha 1	Extracellular matrix remodeling; driver of desmoplastic stroma and tumor invasion [10]
	Hsa.831	*GUCA2A*	Guanylin	Mucosal differentiation hormone; loss is a universal hallmark of early colorectal adenoma [11]
	Hsa.3331	*DES*	Desmin	Smooth muscle intermediate filament; distinguishes mucosal integrity from tumor infiltration [11]
	Hsa.1832	*ACTG2*	Actin gamma 2	Cytoskeletal actin of smooth muscle; delineates muscularis mucosae invasion boundaries [11]
	Hsa.692	*RPS18*	Ribosomal protein S18	40S ribosomal subunit; marker of hyper-activated protein translation in neoplastic cells [11]
	Hsa.37937	*RPL37A*	Ribosomal protein L37a	60S ribosomal subunit protein; reflects enhanced metabolic and translational demand in tumor stroma [11]
Prostate	1005_at	*HPN*	Hepsin	Type II transmembrane serine protease; hallmark diagnostic biomarker overexpressed in prostate cancer [12]
	37639_at	*AMACR*	Alpha-methylacyl-CoA racemase	Peroxisomal enzyme; clinical immunohistochemical gold standard for prostatic adenocarcinoma [13]
	38322_at	*PRAC*	Prostate and colon associated	Small nuclear protein; prostate-specific differentiation marker with restricted normal expression [12]

Section S11: Full Granular Classification Accuracy and Uncertainty Landscape (Table S11)

Table S11 reports the complete classification accuracy performance (Mean ± Std) across 10 independent stochastic runs for all 12 evaluated algorithms across the 20 benchmark datasets. Statistical significance markers ($+/\approx/-$) are derived from pairwise two-sided Mann–Whitney U tests ($\alpha = 0.05$) evaluated directly on the raw 10 independent stochastic runs ($n_1 = n_2 = 10$) on each benchmark dataset, providing full transparency on run-level stability and distributional variance.

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Table S11: Complete classification accuracy (Mean ± Std) across 10 independent runs on all 20 benchmark datasets.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	1.0000±0.0000	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	0.9995±0.0001 $^{+}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$
ALL4	0.9480±0.0068	0.9367±0.0125 $^{\approx}$	0.9356±0.0124 $^{+}$	0.8656±0.0558 $^{+}$	0.9140±0.0124 $^{+}$	0.8580±0.0318 $^{+}$	0.8308±0.0152 $^{+}$	0.7634±0.0088 $^{+}$	0.8250±0.0245 $^{+}$	0.7918±0.0197 $^{+}$	0.7043±0.0136 $^{+}$	0.7846±0.0080 $^{+}$
ALL_AML	1.0000±0.0000	1.0000±0.0000 $^{\approx}$	0.9947±0.0069 $^{\approx}$	0.9857±0.0000 $^{+}$	1.0000±0.0000 $^{\approx}$	0.9914±0.0074 $^{+}$	0.9868±0.0044 $^{+}$	0.9861±0.0000 $^{+}$	0.9857±0.0000 $^{+}$	0.9857±0.0000 $^{+}$	0.9778±0.0072 $^{+}$	0.9871±0.0045 $^{+}$
Adenoma	1.0000±0.0000	1.0000±0.0000 $^{\approx}$	0.9971±0.0090 $^{\approx}$	0.9714±0.0000 $^{+}$	1.0000±0.0000 $^{\approx}$	0.9800±0.0138 $^{+}$	0.9767±0.0121 $^{+}$	0.9722±0.0000 $^{+}$	0.9714±0.0000 $^{+}$	0.9714±0.0000 $^{+}$	0.9611±0.0143 $^{+}$	0.9714±0.0000 $^{+}$
BASEHOCK	0.9045±0.0117	0.8933±0.0081 $^{+}$	0.8627±0.0116 $^{+}$	0.8802±0.0079 $^{+}$	0.9082±0.0044 $^{\approx}$	0.9045±0.0095 $^{\approx}$	0.8739±0.0135 $^{+}$	0.8897±0.0043 $^{+}$	0.9170±0.0074 $^{\approx}$	0.8743±0.0115 $^{+}$	0.8610±0.0094 $^{+}$	0.8994±0.0038 $^{\approx}$
COIL20	1.0000±0.0000	0.9976±0.0010 $^{+}$	0.9955±0.0008 $^{+}$	0.9944±0.0009 $^{+}$	0.9991±0.0007 $^{+}$	0.9980±0.0011 $^{+}$	0.9939±0.0012 $^{+}$	0.9953±0.0005 $^{+}$	0.9988±0.0003 $^{+}$	0.9951±0.0014 $^{+}$	0.9942±0.0009 $^{+}$	0.9967±0.0006 $^{+}$
Colon	0.9452±0.0082	0.9294±0.0107 $^{+}$	0.9182±0.0079 $^{+}$	0.8777±0.0193 $^{+}$	0.9468±0.0078 $^{\approx}$	0.9268±0.0167 $^{+}$	0.8914±0.0103 $^{+}$	0.8694±0.0051 $^{+}$	0.9086±0.0126 $^{+}$	0.8733±0.0114 $^{+}$	0.8048±0.0208 $^{+}$	0.8885±0.0073 $^{+}$
DLBCL	0.9999±0.0002	1.0000±0.0000 $^{\approx}$	0.9987±0.0042 $^{\approx}$	0.9859±0.0074 $^{+}$	1.0000±0.0000 $^{\approx}$	0.9935±0.0092 $^{\approx}$	0.9966±0.0056 $^{+}$	0.9883±0.0041 $^{+}$	0.9924±0.0065 $^{+}$	0.9884±0.0073 $^{+}$	0.9675±0.0110 $^{+}$	0.9899±0.0081 $^{+}$
Gastric	0.9763±0.0038	0.9646±0.0104 $^{+}$	0.9723±0.0097 $^{\approx}$	0.9015±0.0649 $^{+}$	0.9569±0.0159 $^{+}$	0.9154±0.0334 $^{+}$	0.8886±0.0097 $^{+}$	0.8231±0.0081 $^{+}$	0.8385±0.0208 $^{+}$	0.8646±0.0097 $^{+}$	0.7769±0.0181 $^{+}$	0.8369±0.0108 $^{+}$
Gastric1	0.9779±0.0002	0.9719±0.0000 $^{+}$	0.9636±0.0029 $^{+}$	0.9602±0.0047 $^{+}$	0.9722±0.0000 $^{+}$	0.9623±0.0036 $^{+}$	0.9588±0.0029 $^{+}$	0.9583±0.0000 $^{+}$	0.9623±0.0036 $^{+}$	0.9588±0.0022 $^{+}$	0.9500±0.0072 $^{+}$	0.9581±0.0000 $^{+}$
Gastric2	0.9982±0.0030	0.9936±0.0034 $^{+}$	0.9880±0.0042 $^{+}$	0.9822±0.0075 $^{+}$	0.9952±0.0042 $^{\approx}$	0.9887±0.0042 $^{+}$	0.9830±0.0002 $^{+}$	0.9823±0.0034 $^{+}$	0.9815±0.0040 $^{+}$	0.9837±0.0000 $^{+}$	0.9750±0.0026 $^{+}$	0.9815±0.0040 $^{+}$
ISOLET	0.8880±0.0084	0.8955±0.0064 $^{-}$	0.8801±0.0086 $^{+}$	0.8783±0.0066 $^{+}$	0.9112±0.0080 $^{-}$	0.9000±0.0064 $^{-}$	0.8775±0.0123 $^{\approx}$	0.8763±0.0043 $^{+}$	0.9164±0.0082 $^{-}$	0.8648±0.0089 $^{+}$	0.8760±0.0056 $^{+}$	0.8942±0.0031 $^{-}$
Leukaemia	1.0000±0.0000	1.0000±0.0000 $^{\approx}$	0.9960±0.0064 $^{\approx}$	0.9857±0.0000 $^{+}$	1.0000±0.0000 $^{\approx}$	0.9957±0.0069 $^{\approx}$	0.9895±0.0068 $^{+}$	0.9861±0.0000 $^{+}$	0.9857±0.0000 $^{+}$	0.9871±0.0045 $^{+}$	0.9764±0.0094 $^{+}$	0.9857±0.0000 $^{+}$
Lymphoma	1.0000±0.0000	1.0000±0.0000 $^{\approx}$	0.9867±0.0115 $^{+}$	0.8978±0.0155 $^{+}$	0.9978±0.0070 $^{\approx}$	0.9422±0.0318 $^{+}$	0.9263±0.0108 $^{+}$	0.8889±0.0000 $^{+}$	0.9044±0.0107 $^{+}$	0.8978±0.0155 $^{+}$	0.8400±0.0230 $^{+}$	0.8933±0.0094 $^{+}$
Myeloma	0.9083±0.0112	0.8889±0.0089 $^{+}$	0.8710±0.0086 $^{+}$	0.8466±0.0092 $^{+}$	0.8884±0.0098 $^{+}$	0.8722±0.0063 $^{+}$	0.8552±0.0089 $^{+}$	0.8387±0.0088 $^{+}$	0.8757±0.0078 $^{+}$	0.8367±0.0087 $^{+}$	0.8012±0.0128 $^{+}$	0.8536±0.0062 $^{+}$
PCMAC	0.8592±0.0126	0.8360±0.0077 $^{+}$	0.8000±0.0113 $^{+}$	0.8085±0.0085 $^{+}$	0.8410±0.0040 $^{+}$	0.8301±0.0152 $^{+}$	0.8038±0.0103 $^{+}$	0.8161±0.0060 $^{+}$	0.8498±0.0113 $^{\approx}$	0.7985±0.0139 $^{+}$	0.7601±0.0187 $^{+}$	0.8310±0.0065 $^{+}$
Prostate	0.9670±0.0036	0.9552±0.0050 $^{+}$	0.9600±0.0055 $^{+}$	0.8732±0.0092 $^{+}$	0.9353±0.0069 $^{+}$	0.8930±0.0114 $^{+}$	0.8715±0.0097 $^{+}$	0.8686±0.0051 $^{+}$	0.8940±0.0109 $^{+}$	0.8603±0.0113 $^{+}$	0.8235±0.0092 $^{+}$	0.8863±0.0048 $^{+}$
SRBCT	1.0000±0.0000	1.0000±0.0000 $^{\approx}$	1.0000±0.0000 $^{\approx}$	0.9682±0.0107 $^{+}$	1.0000±0.0000 $^{\approx}$	0.9865±0.0146 $^{+}$	0.9837±0.0057 $^{+}$	0.9663±0.0051 $^{+}$	0.9976±0.0050 $^{\approx}$	0.9709±0.0130 $^{+}$	0.9325±0.0236 $^{+}$	0.9953±0.0061 $^{\approx}$
T1D	0.8449±0.0105	0.8043±0.0139 $^{+}$	0.8054±0.0094 $^{+}$	0.7558±0.0313 $^{+}$	0.7752±0.0115 $^{+}$	0.7446±0.0156 $^{+}$	0.7204±0.0073 $^{+}$	0.7178±0.0052 $^{+}$	0.7296±0.0061 $^{+}$	0.7060±0.0070 $^{+}$	0.6723±0.0109 $^{+}$	0.7285±0.0047 $^{+}$
madelon	0.8731±0.0188	0.8567±0.0231 $^{\approx}$	0.7710±0.0168 $^{+}$	0.7667±0.0124 $^{+}$	0.8677±0.0169 $^{\approx}$	0.8008±0.0166 $^{+}$	0.7621±0.0224 $^{+}$	0.7598±0.0065 $^{+}$	0.8255±0.0082 $^{+}$	0.7599±0.0087 $^{+}$	0.7695±0.0123 $^{+}$	0.7882±0.0059 $^{+}$
Average	**0.9545**	**0.9462**	**0.9348**	**0.9093**	**0.9454**	**0.9242**	**0.9085**	**0.8973**	**0.9180**	**0.8985**	**0.8712**	**0.9075**
$+/\approx/-$	—	11/8/1	13/7/0	19/1/0	8/11/1	15/4/1	19/1/0	19/1/0	15/3/2	19/1/0	19/1/0	17/2/1

Note: Values report mean classification accuracy ± standard deviation across 10 independent stochastic runs. Superscripts $+/\approx/-$ denote that KDS-EL is statistically superior, equivalent, or inferior to the corresponding competitor method under pairwise two-sided Mann–Whitney U tests ($\alpha = 0.05$) evaluated directly on the raw 10 independent runs ($n_1 = n_2 = 10$) on each benchmark dataset. The final rows report unweighted cross-dataset average accuracies and overall pairwise win/tie/loss tallies against KDS-EL.